Track- Duality AI's Offroad Semantic Scene Segmentation

TerrainIQ- Smart Offroad Vision

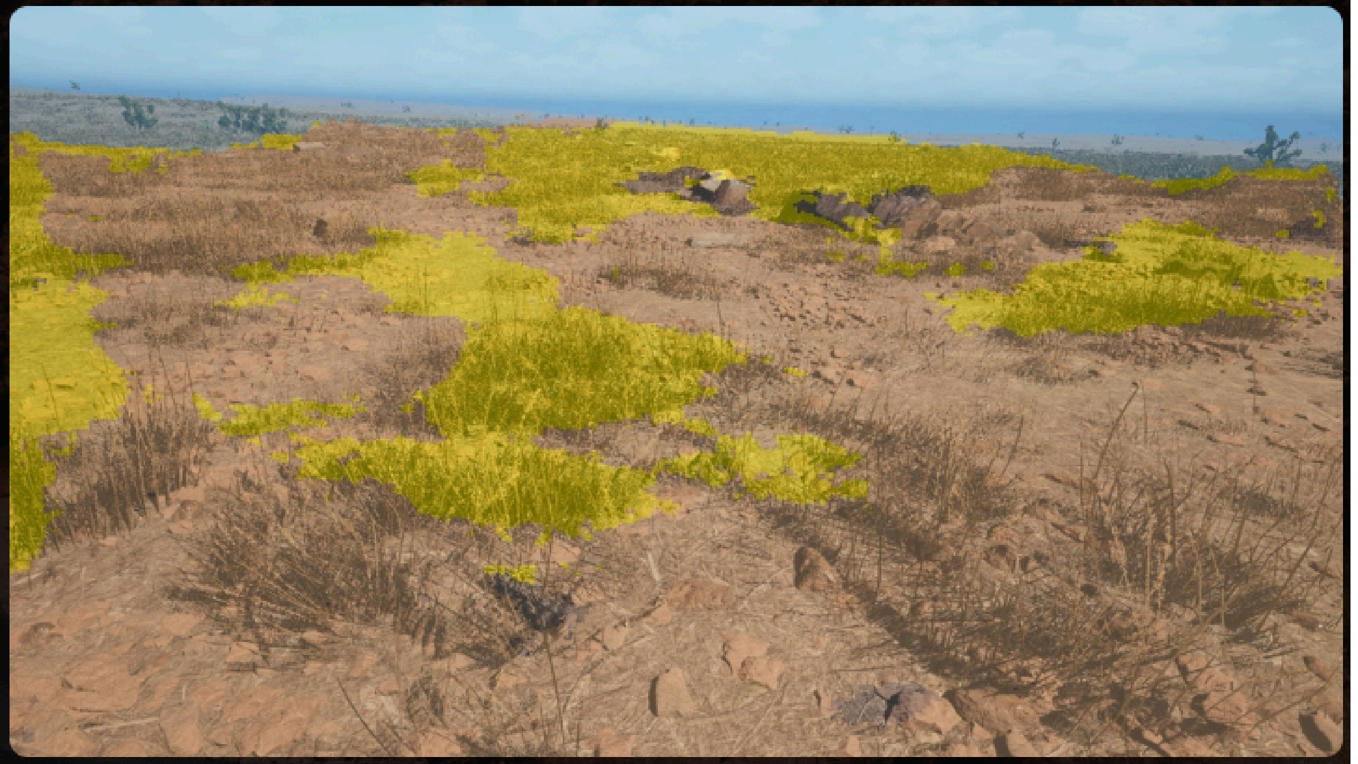
~Turning Unseen Terrain into Intelligent Decisions

Team Code Crafters:
Sakshi Mittal
Saumya Dwivedi

Live Demo:
terrainiq.streamlit.app

GitHub:
https://github.com/saumyadwiv/TerrainIQ


Original Terrain


Segmented Terrain Map

Problem

Offroad Autonomy Needs Terrain Intelligence

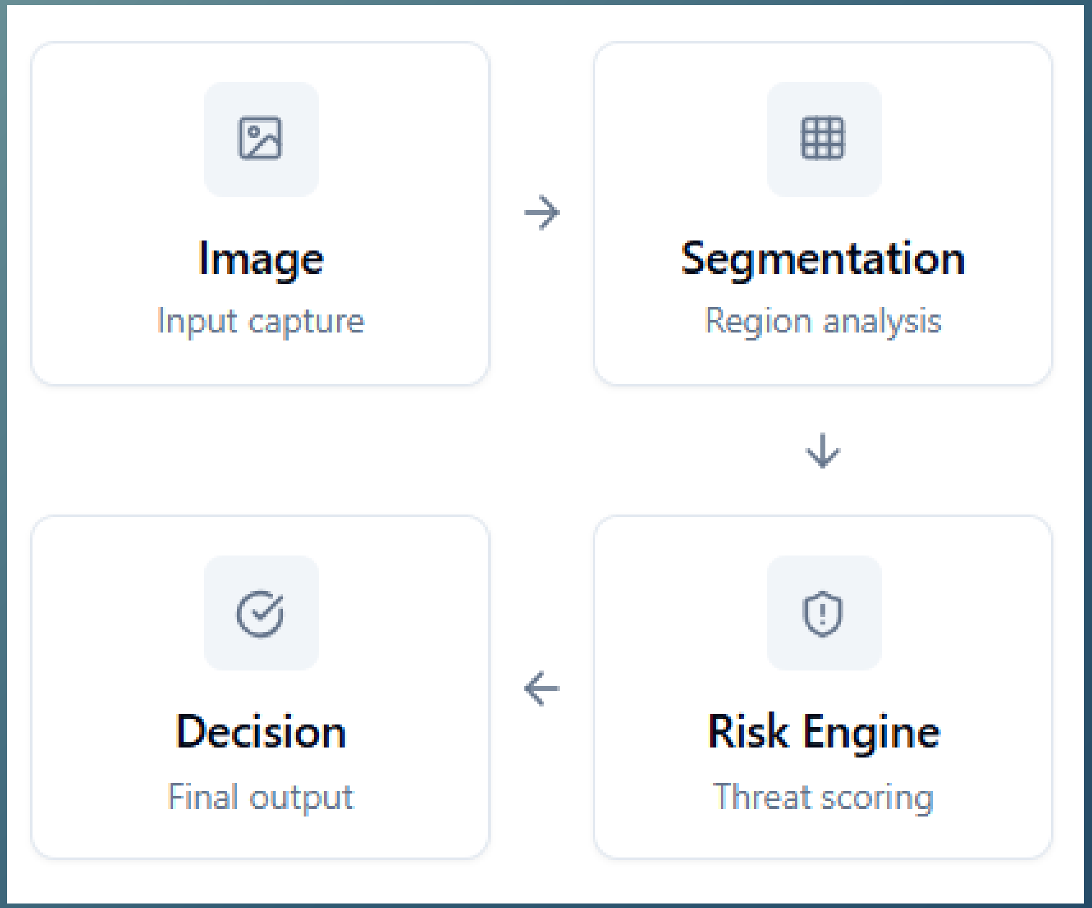
Autonomous and emergency vehicles require:
1. Accurate terrain understanding
2. Obstacle awareness
3. Safe route decisions

But real-world data has problems:
1. Expensive to collect
2. Difficult to annotate
3. Limited environments

The hackathon challenge:
Train a segmentation model using synthetic desert environments and generalize to new terrain.

Our Innovation

End-to-End Pipeline



TerrainIQ transforms raw terrain images into actionable operational intelligence.

TerrainIQ – From Segmentation to Intelligence

Most solutions stop at semantic segmentation.
TerrainIQ goes further. We convert terrain images into actionable mobility intelligence for real-world operations.
Other teams built segmentation models.
We built an intelligent terrain decision system.

What TerrainIQ Does

01 Pixel-Level Terrain Segmentation

02 Terrain Risk Scoring

03 Vehicle Compatibility Analysis

04 Operational Decision Support

System Architecture

AI Architecture

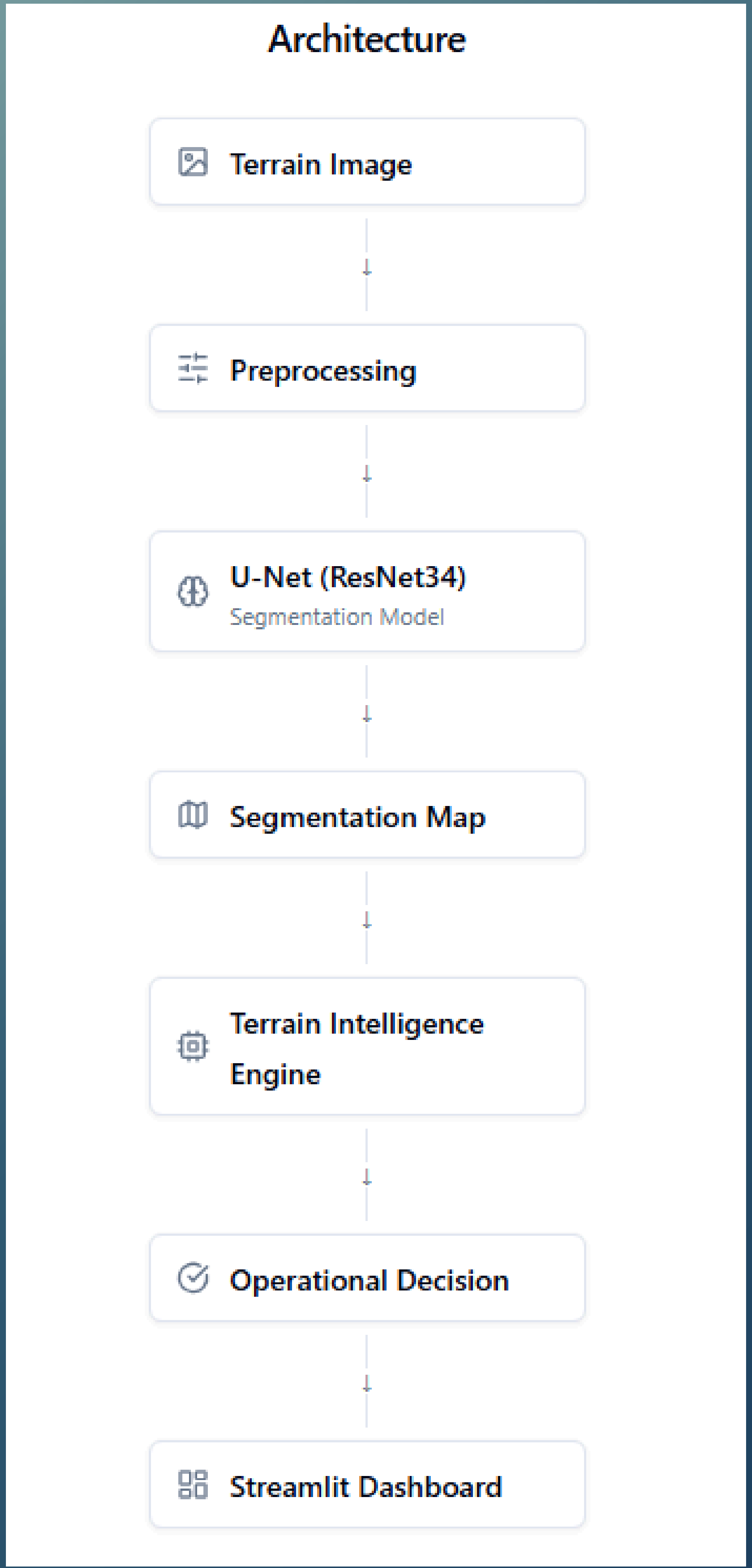
Deep Learning Model
U-Net with ResNet34 Encoder

Why we chose this model:

- High accuracy
- Lightweight
- Fast inference
- Proven segmentation architecture

10 Classes:

- Trees
- Bushes
- Rocks
- Ground
- Sky
- Logs
- Flowers
- Grass
- Clutter



Results

Model Performance include:

Metrics

Pixel-wise Segmentation

- Validation IoU: 0.39001905213330344
- Cross-Environment IoU: 0.23897219365834986
- IoU per class
- Average inference time: 0.009516897201538086 seconds
- Average inference time per image: 0.0009420070343626712
- FPS: 1061.563198067374
- Confusion matrix

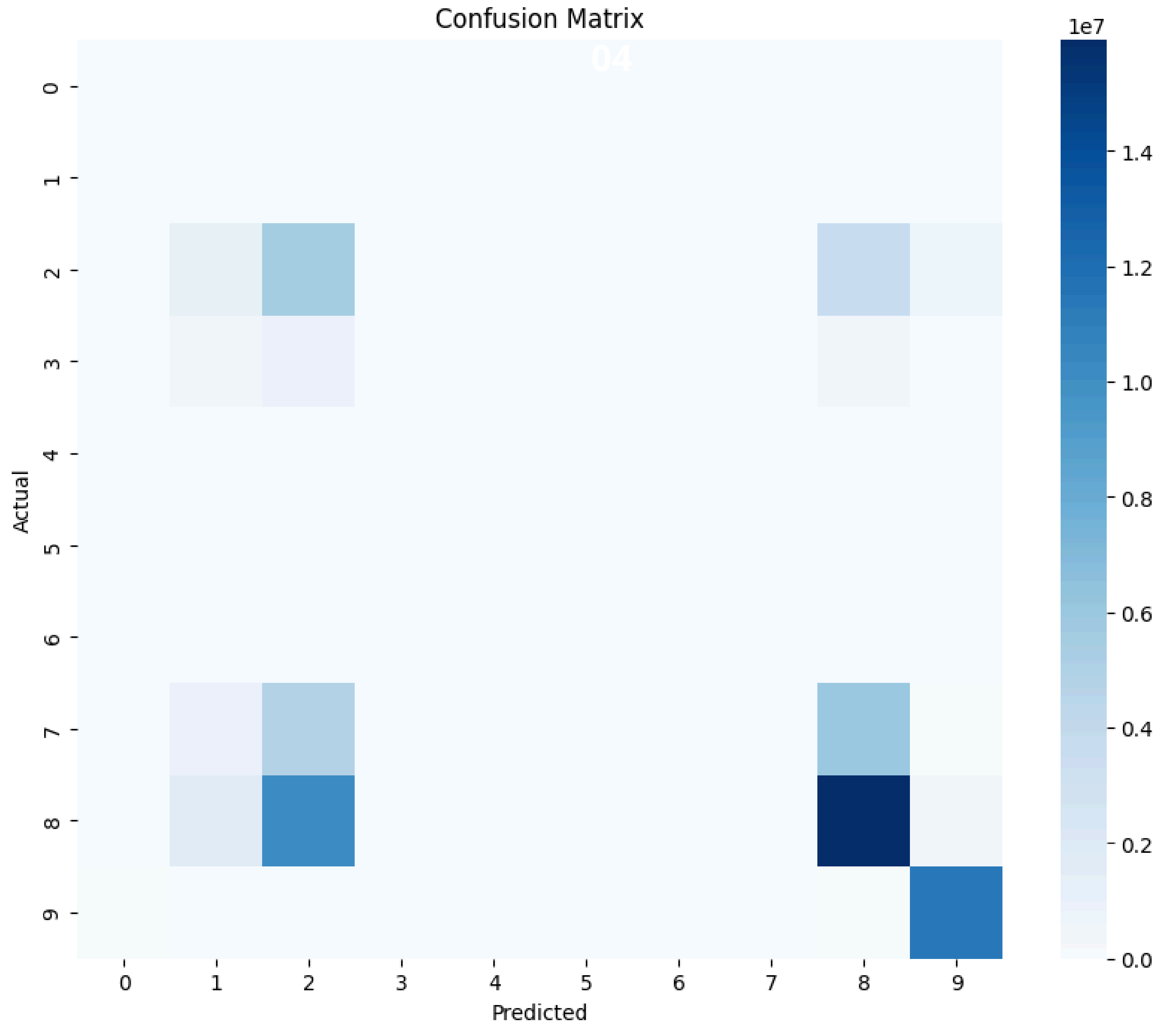
```
100%|██████████| 126/126 [00:44<00:00,  2.82it/s]Class 0 IoU: 0.0182210913789
Class 1 IoU: 3.3000947787213187e-06
Class 2 IoU: 0.20544026995793593
Class 3 IoU: 0.00913417412697082
Class 4 IoU: 0.0
Class 5 IoU: 0.0
Class 6 IoU: 0.0
Class 7 IoU: 0.004225262195999372
Class 8 IoU: 0.41291571306916
Class 9 IoU: 0.8707744831046752
```

Generalization

Tested on unseen terrain images.

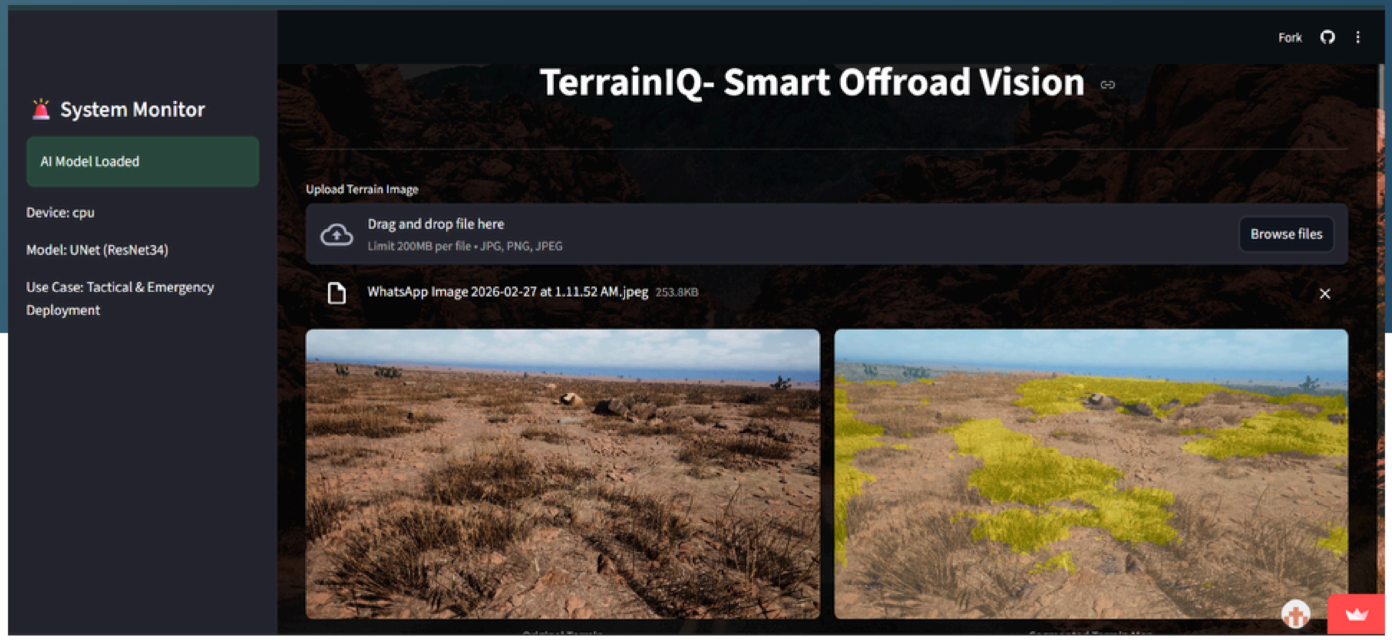
Results:

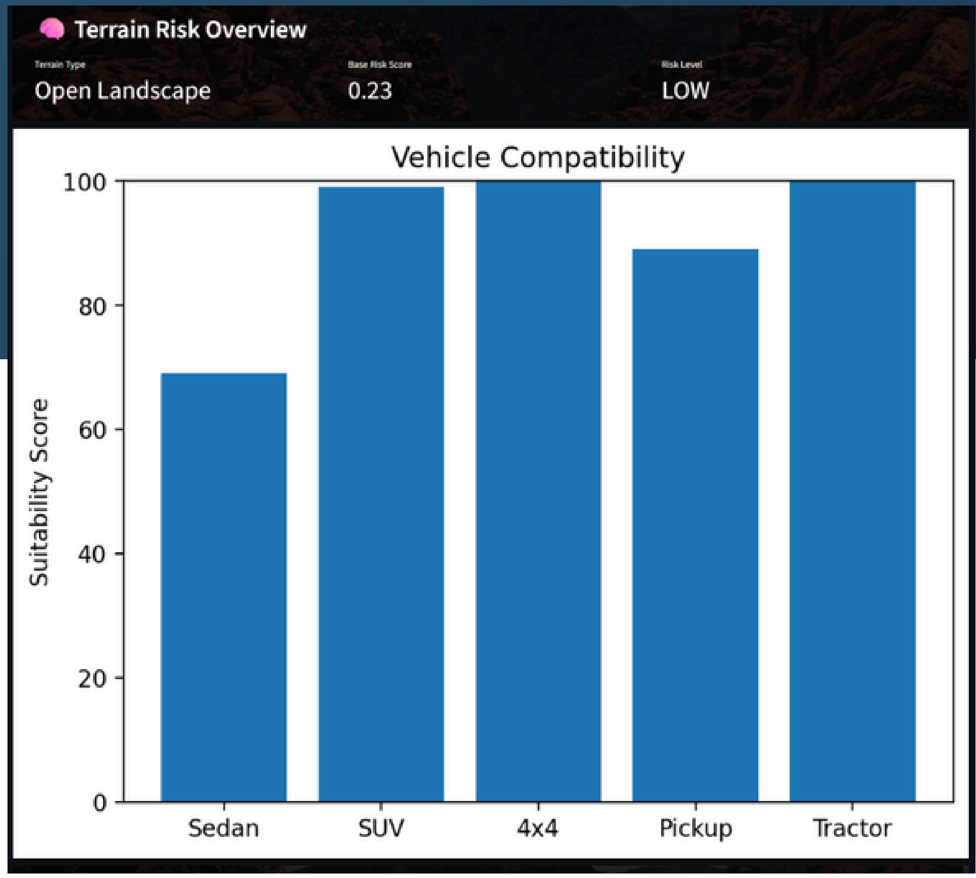
1. Clear terrain boundaries
2. Accurate vegetation detection
3. Strong ground detection

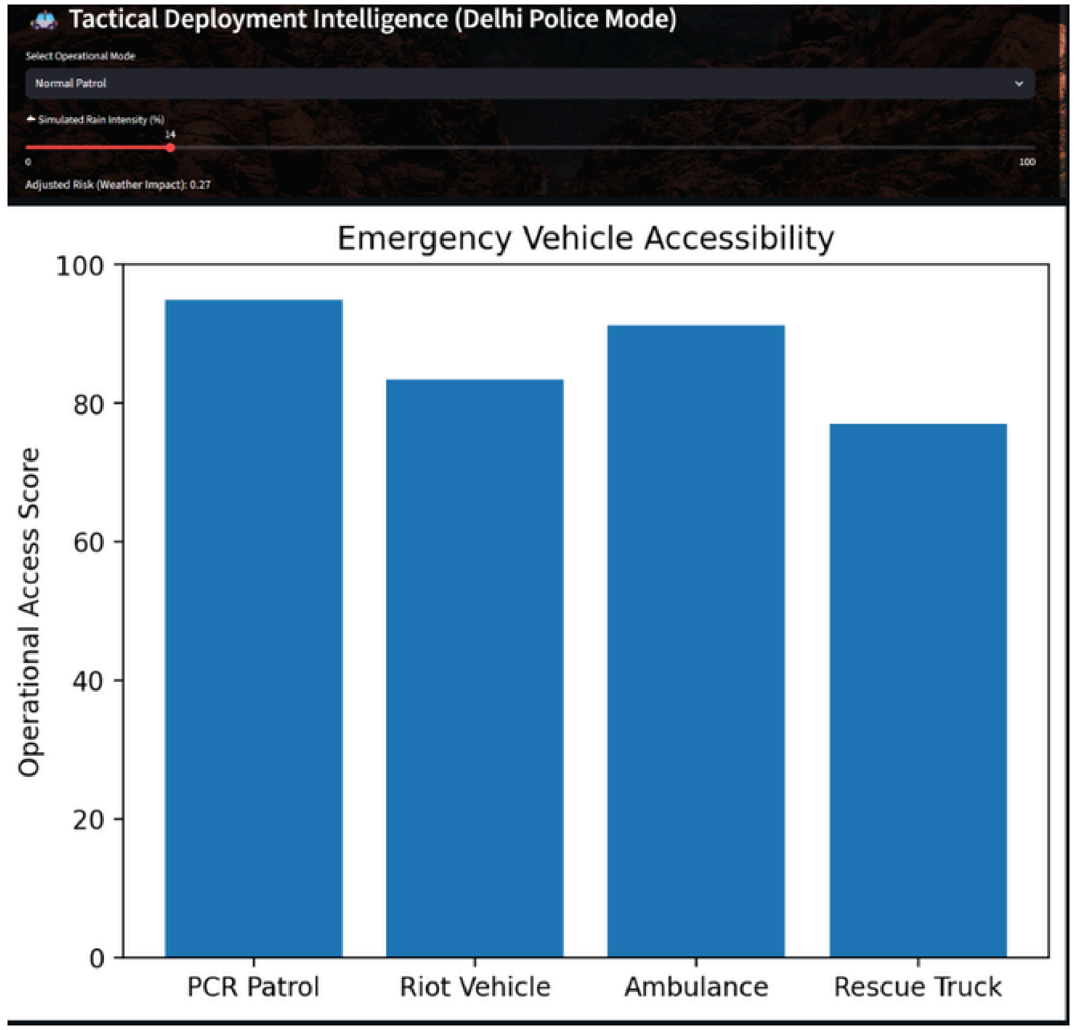


Live AI System

TerrainIQ Smart Dashboard







Workflow

1 . **Upload Terrain Image**
The user uploads an off-road terrain image through the TerrainIQ dashboard.

2 . **AI Terrain Segmentation**
The deep learning model (U-Net with ResNet34 encoder) analyzes the image and generates a pixel-level terrain map.

3 .**Terrain Intelligence Analysis**
The Terrain Intelligence Engine calculates terrain risk, vehicle suitability, and operational feasibility.

4 . **Decision Output**
The system provides risk scores, deployment insights, and a final operational recommendation through the dashboard.

Features

✓ Terrain Risk Overview
Evaluates overall terrain safety by analyzing surface conditions and obstacle density.

✓ Vehicle Compatibility
Determines which vehicles can safely operate on the terrain based on mobility suitability scores.

✓ Emergency Deployment
Simulates whether emergency vehicles can access the terrain quickly and safely.

✓ Drone Index
Estimates how effective drone surveillance would be based on terrain openness and obstacles.

Technical Innovation

Key Innovations

1 — Intelligent Risk Engine

Terrain segmentation converted into:

- Mobility scores
- Operational safety

2 — Tactical Deployment Simulation

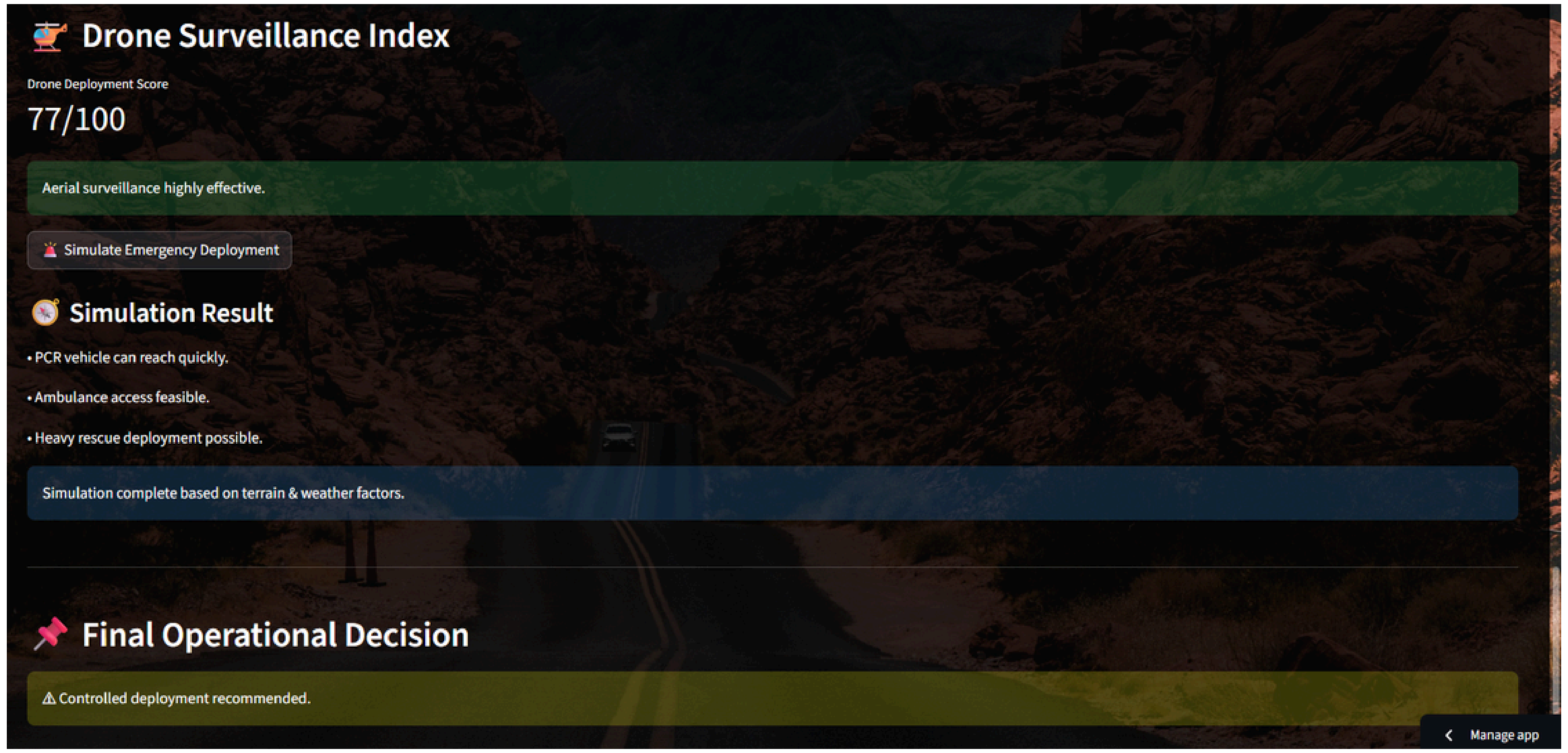
Predicts:

- Emergency vehicle access
- Rescue feasibility

3 — Real Deployment

- Live web app
- Reproducible model
- Interactive dashboard

This screams production-ready AI.



Original Terrain

Segmented Terrain Map

Impact

Real World Impact

TerrainIQ enables:

- Emergency response planning
- Law enforcement mobility
- Agricultural robotics
- Autonomous rovers

Domain	How TerrainIQ Helps
Emergency Response	Faster rescue planning
Autonomous Vehicles	Safer navigation decisions
Agriculture Robotics	Terrain-aware operations
Disaster Management	Accessibility analysis
Defense & Surveillance	Mission planning support

Future Work

Future Work

- Real-time drone feeds
- Video segmentation
- Edge AI deployment
- Path planning integration

~TerrainIQ transforms terrain perception into intelligent autonomous decisions.

THANK YOU